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METHOD FOR PROVIDING FILTER AND ELECTRONIC DEVICE SUPPORTING THE SAME

Abstract

An electronic device is provided. The electronic device includes a display, a processor functionally connected with the display, and a memory functionally connected with the processor. The memory stores instructions configured to, when executed, enable the processor to display a first image through the display, display one or more second images through the display while displaying the first image, select a third image from among the one or more second images, identify a value of at least one property of the third image, generate a filter for applying the value of the at least one property to an image, apply the value of the at least one property to the first image using the filter, display the first image, to which the value of the at least one property is applied, through the display, and store the filter in the memory.

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Background/Summary

[0001] CROSS-REFERENCE TO RELATED APPLICATION(S) [0002] This application is a Continuation application of prior application Ser. No. 18/305,886, filed on Apr. 24, 2023, which is a continuation application of prior application Ser. No. 17/168,806, filed on Feb. 5, 2021, which has issued as U.S. Pat. No. 11,662,898 on May 30, 2023, which is based on and claims priority under 35 U.S.C. § 119 (a) of a Korean patent application number 10-2020-0014431, filed on Feb. 6, 2020, in the Korean Intellectual Property Office, the disclosure of which is incorporated by reference herein in its entirety.

BACKGROUND

1. Field

[0003] The disclosure relates to methods for providing a filter and electronic devices supporting the same.

2. Description of Related Art

[0004] Electronic devices such as smart phones provide an image editing function. For example, an electronic device allows a user to edit images acquired through a camera or stored in a memory using an image editing program.

[0005] Recently, electronic devices have provided a filter function for changing the properties of an image (e.g., an original image). For example, while displaying an image, the electronic device may display images (e.g., thumbnail images) for a plurality of filters individually corresponding to a plurality of properties of the image. When one image is selected by a user from among the images for the plurality of filters, the electronic device may provide an image to which an image property corresponding to the filter of the selected image has been applied.

[0006] The above information is presented as background information only to assist with an understanding of the disclosure. No determination has been made, and no assertion is made, as to whether any of the above might be applicable as prior art with regard to the disclosure.

SUMMARY

[0007] Aspects of the disclosure are to address at least the above-mentioned problems and/or disadvantages and to provide at least the advantages described below. Accordingly, an aspect of the disclosure is to provide the electronic device for providing only filters (or filter functions) generated by the generator (or developer) of the camera application, image application (e.g., Gallery application), or an image editing application (e.g., an application for providing a photo

editor tool) Accordingly, filters (e.g., types of filters) that the electronic device may provide to the user may be limited.

[0008] Another aspect of the disclosure is to provide methods for providing a filter and electronic devices for supporting the same, which may provide a filter for the user to obtain images having a desired atmosphere, based on the user's input (or selection).

[0009] Objects of the disclosure are not limited to the foregoing, and other unmentioned objects would be apparent to one of ordinary skill in the art from the following description.

[0010] Additional aspects will be set forth in part in the description which follows and, in part, will be apparent from the description, or may be learned by practice of the presented embodiments.

[0011] In accordance with an aspect of the disclosure, an electronic device is provided. The electronic device includes a display, a processor functionally connected with the display, and a memory functionally connected with the processor. The memory stores instructions configured to, when executed, enable the processor to display a first image through the display, display one or more second images through the display while displaying the first image, select a third image from among the one or more second images, identify a value of at least one property of the third image, generate a filter for applying the value of the at least one property to an image, apply the value of the at least one property to the first image using the filter, display the first image, to which the value of the at least one property is applied, through the display, and store the filter in the memory.

[0012] In accordance with another aspect of the disclosure, a method is provided. The method includes displaying a first image through a display of an electronic device, displaying one or more second images through the display while displaying the first image, selecting a third image from among the one or more second images, identifying a value of at least one property of the third image, generating a filter for applying the value of the at least one property to an image, applying the value of the at least one property to the first image using the filter, displaying the first image, to which the value of the at least one property is applied, through the display, and storing the filter in a memory of the electronic device.

[0013] In accordance with another aspect of the disclosure, an electronic device is provided. The electronic device includes a camera, a display, a processor functionally connected with the camera and the display, and a memory functionally connected with the processor. The memory stores instructions configured to, when executed, enable the processor to display a first image input in real-time using the camera and images indicating one or more filters through the display, display one or more second images through the display, based on a user input for generating a filter while displaying the first image and the images, select a third image from among the one or more second images, identify a value of at least one property of the third image, generate a filter for applying the value of the at least one property to an image, apply the value of the at least one property to the first image using the filter, display the first image, to which the value of the at least one property is applied, through the display, store the filter in the memory, and display an image indicating the filter and the images through the display.

[0014] According to various embodiments, the method for providing a filter and the electronic devices for supporting the same may provide a filter for the user to obtain images having a desired atmosphere, based on the user's input (or selection).

[0015] Other aspects, advantages, and salient features of the disclosure will become apparent to those skilled in the art from the following detailed description, which, taken in conjunction with the annexed drawings, discloses various embodiments of the disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0016] The above and other aspects, features, and advantages of certain embodiments of the

disclosure will be more apparent from the following description taken in conjunction with the accompanying drawings, in which:

[0017] FIG. 1 is a block diagram illustrating an electronic device in a network environment according to an embodiment of the disclosure;

[0018] FIG. 2 is a flowchart illustrating a method for providing a filter in an electronic device according to an embodiment of the disclosure;

[0019] FIG. 3 is a flowchart illustrating a method for identifying at least one property value of a third image, according to an embodiment of the disclosure;

[0020] FIG. 4 is a flowchart illustrating a method for identifying at least one property value by recognizing an object recognized in a third image, according to an embodiment of the disclosure;

[0021] FIG. 5 is a flowchart illustrating a method for adjusting at least one property value based on a user interface, according to an embodiment of the disclosure;

[0022] FIG. 6 is a flowchart illustrating a method for adjusting at least one property value based on a plurality of images, according to an embodiment of the disclosure;

[0023] FIG. 7 is a flowchart illustrating a method for generating a filter based on a portion of a third image designated based on a user input, according to an embodiment of the disclosure;

[0024] FIG. 8 is a flowchart illustrating a method for providing a filter using a camera application according to an embodiment of the disclosure;

[0025] FIG. 9 is a flowchart illustrating a method for providing a filter using an image application according to an embodiment of the disclosure;

[0026] FIGS. 10A and 10B are views illustrating a method for providing a filter using a camera application according to various embodiments of the disclosure;

[0027] FIGS. 11A and 11B are views illustrating a method for providing a filter using an image application according to various embodiments of the disclosure;

[0028] FIG. 12 is a view illustrating an example method for identifying at least one property value of a third image, according to an embodiment of the disclosure;

[0029] FIG. 13 is a view illustrating an example method for identifying at least one property value by recognizing an object recognized in a third image, according to an embodiment of the disclosure;

[0030] FIG. 14 is a view illustrating an example method for adjusting at least one property value based on a user interface, according to an embodiment of the disclosure; and

[0031] FIG. 15 is a view illustrating an example method for adjusting at least one property value based on a plurality of images, according to an embodiment of the disclosure.

[0032] Throughout the drawings, like reference numerals will be understood to refer to like parts, components, and structures.

DETAILED DESCRIPTION

[0033] The following description with reference to the accompanying drawings is provided to assist in a comprehensive understanding of various embodiments of the disclosure as defined by the claims and their equivalents. It includes various specific details to assist in that understanding, but these are to be regarded as merely exemplary. Accordingly, those of ordinary skill in the art will recognize that various changes and modifications of the various embodiments described herein can be made without departing from the scope and spirit of the disclosure. In addition, descriptions of well-known functions and constructions may be omitted for clarity and conciseness.

[0034] The terms and words used in the following description and claims are not limited to the bibliographical meanings, but are merely used by the inventor to enable a clear and consistent understanding of the disclosure. Accordingly, it should be apparent to those skilled in the art that the following description of various embodiments of the disclosure is provided for illustration purposes only and not for the purpose of limiting the disclosure as defined by the appended claims and their equivalents.

[0035] It is to be understood that the singular forms “a,” “an,” and “the” include plural referents

unless the context clearly dictates otherwise. Thus, for example, reference to “a component surface” includes reference to one or more of such surfaces.

[0036] FIG. 1 is a block diagram illustrating an electronic device **101** in a network environment **100** according to an embodiment of the disclosure.

[0037] Referring to FIG. 1, the electronic device **101** in the network environment **100** may communicate with an electronic device **102** via a first network **198** (e.g., a short-range wireless communication network), or an electronic device **104** or a server **108** via a second network **199** (e.g., a long-range wireless communication network). According to an embodiment, the electronic device **101** may communicate with the electronic device **104** via the server **108**. According to an embodiment, the electronic device **101** may include a processor **120**, memory **130**, an input device **150**, a sound output device **155**, a display device **160**, an audio module **170**, a sensor module **176**, an interface **177**, a haptic module **179**, a camera module **180**, a power management module **188**, a battery **189**, a communication module **190**, a subscriber identification module (SIM) **196**, or an antenna module **197**. In some embodiments, at least one (e.g., the display device **160** or the camera module **180**) of the components may be omitted from the electronic device **101**, or one or more other components may be added in the electronic device **101**. In some embodiments, some of the components may be implemented as single integrated circuitry. For example, the sensor module **176** (e.g., a fingerprint sensor, an iris sensor, or an illuminance sensor) may be implemented as embedded in the display device **160** (e.g., a display).

[0038] The processor **120** may execute, for example, software (e.g., a program **140**) to control at least one other component (e.g., a hardware or software component) of the electronic device **101** coupled with the processor **120**, and may perform various data processing or computation. According to one embodiment, as at least part of the data processing or computation, the processor **120** may load a command or data received from another component (e.g., the sensor module **176** or the communication module **190**) in volatile memory **132**, process the command or the data stored in the volatile memory **132**, and store resulting data in non-volatile memory **134**. According to an embodiment, the processor **120** may include a main processor **121** (e.g., a central processing unit (CPU) or an application processor (AP)), and an auxiliary processor **123** (e.g., a graphics processing unit (GPU), an image signal processor (ISP), a sensor hub processor, or a communication processor (CP)) that is operable independently from, or in conjunction with, the main processor **121**. Additionally or alternatively, the auxiliary processor **123** may be adapted to consume less power than the main processor **121**, or to be specific to a specified function. The auxiliary processor **123** may be implemented as separate from, or as part of the main processor **121**.

[0039] The auxiliary processor **123** may control at least some of functions or states related to at least one component (e.g., the display device **160**, the sensor module **176**, or the communication module **190**) among the components of the electronic device **101**, instead of the main processor **121** while the main processor **121** is in an inactive (e.g., sleep) state, or together with the main processor **121** while the main processor **121** is in an active state (e.g., executing an application). According to an embodiment, the auxiliary processor **123** (e.g., an image signal processor or a communication processor) may be implemented as part of another component (e.g., the camera module **180** or the communication module **190**) functionally related to the auxiliary processor **123**.

[0040] The memory **130** may store various data used by at least one component (e.g., the processor **120** or the sensor module **176**) of the electronic device **101**. The various data may include, for example, software (e.g., the program **140**) and input data or output data for a command related thereto. The memory **130** may include the volatile memory **132** or the non-volatile memory **134**.

[0041] The program **140** may be stored in the memory **130** as software, and may include, for example, an operating system (OS) **142**, middleware **144**, or an application **146**.

[0042] The input device **150** may receive a command or data to be used by other component (e.g., the processor **120**) of the electronic device **101**, from the outside (e.g., a user) of the electronic device **101**. The input device **150** may include, for example, a microphone, a mouse, a keyboard, or

a digital pen (e.g., a stylus pen).

[0043] The sound output device **155** may output sound signals to the outside of the electronic device **101**. The sound output device **155** may include, for example, a speaker or a receiver. The speaker may be used for general purposes, such as playing multimedia or playing record, and the receiver may be used for an incoming calls. According to an embodiment, the receiver may be implemented as separate from, or as part of the speaker.

[0044] The display device **160** may visually provide information to the outside (e.g., a user) of the electronic device **101**. The display device **160** may include, for example, a display, a hologram device, or a projector and control circuitry to control a corresponding one of the display, hologram device, and projector. According to an embodiment, the display device **160** may include touch circuitry adapted to detect a touch, or sensor circuitry (e.g., a pressure sensor) adapted to measure the intensity of force incurred by the touch.

[0045] The audio module **170** may convert a sound into an electrical signal and vice versa. According to an embodiment, the audio module **170** may obtain the sound via the input device **150**, or output the sound via the sound output device **155** or a headphone of an external electronic device (e.g., an electronic device **102**) directly (e.g., wiredly) or wirelessly coupled with the electronic device **101**.

[0046] The sensor module **176** may detect an operational state (e.g., power or temperature) of the electronic device **101** or an environmental state (e.g., a state of a user) external to the electronic device **101**, and then generate an electrical signal or data value corresponding to the detected state. According to an embodiment, the sensor module **176** may include, for example, a gesture sensor, a gyro sensor, an atmospheric pressure sensor, a magnetic sensor, an acceleration sensor, a grip sensor, a proximity sensor, a color sensor, an infrared (IR) sensor, a biometric sensor, a temperature sensor, a humidity sensor, or an illuminance sensor.

[0047] The interface **177** may support one or more specified protocols to be used for the electronic device **101** to be coupled with the external electronic device (e.g., the electronic device **102**) directly (e.g., wiredly) or wirelessly. According to an embodiment, the interface **177** may include, for example, a high definition multimedia interface (HDMI), a universal serial bus (USB) interface, a secure digital (SD) card interface, or an audio interface.

[0048] A connecting terminal **178** may include a connector via which the electronic device **101** may be physically connected with the external electronic device (e.g., the electronic device **102**). According to an embodiment, the connecting terminal **178** may include, for example, a HDMI connector, a USB connector, a SD card connector, or an audio connector (e.g., a headphone connector).

[0049] The haptic module **179** may convert an electrical signal into a mechanical stimulus (e.g., a vibration or motion) or electrical stimulus which may be recognized by a user via his tactile sensation or kinesthetic sensation. According to an embodiment, the haptic module **179** may include, for example, a motor, a piezoelectric element, or an electric stimulator.

[0050] The camera module **180** may capture an image or moving images. According to an embodiment, the camera module **180** may include one or more lenses, image sensors, image signal processors, or flashes.

[0051] The power management module **188** may manage power supplied to the electronic device **101**. According to one embodiment, the power management module **388** may be implemented as at least part of, for example, a power management integrated circuit (PMIC).

[0052] The battery **189** may supply power to at least one component of the electronic device **101**. According to an embodiment, the battery **189** may include, for example, a primary cell which is not rechargeable, a secondary cell which is rechargeable, or a fuel cell.

[0053] The communication module **190** may support establishing a direct (e.g., wired) communication channel or a wireless communication channel between the electronic device **101** and the external electronic device (e.g., the electronic device **102**, the electronic device **104**, or the

server **108**) and performing communication via the established communication channel. The communication module **190** may include one or more communication processors that are operable independently from the processor **120** (e.g., the application processor (AP)) and supports a direct (e.g., wired) communication or a wireless communication. According to an embodiment, the communication module **190** may include a wireless communication module **192** (e.g., a cellular communication module, a short-range wireless communication module, or a global navigation satellite system (GNSS) communication module) or a wired communication module **194** (e.g., a local area network (LAN) communication module or a power line communication (PLC) module). A corresponding one of these communication modules may communicate with the external electronic device via the first network **198** (e.g., a short-range communication network, such as Bluetooth™, wireless-fidelity (Wi-Fi) direct, or infrared data association (IrDA)) or the second network **199** (e.g., a long-range communication network, such as a cellular network, the Internet, or a computer network (e.g., LAN or wide area network (WAN))). These various types of communication modules may be implemented as a single component (e.g., a single chip), or may be implemented as multi components (e.g., multi chips) separate from each other. The wireless communication module **192** may identify and authenticate the electronic device **101** in a communication network, such as the first network **198** or the second network **199**, using subscriber information (e.g., international mobile subscriber identity (IMSI)) stored in the subscriber identification module **196**.

[0054] The antenna module **197** may transmit or receive a signal or power to or from the outside (e.g., the external electronic device). According to an embodiment, the antenna module may include one antenna including a radiator formed of a conductor or conductive pattern formed on a substrate (e.g., a printed circuit board (PCB)). According to an embodiment, the antenna module **197** may include a plurality of antennas. In this case, at least one antenna appropriate for a communication scheme used in a communication network, such as the first network **198** or the second network **199**, may be selected from the plurality of antennas by, e.g., the communication module **190**. The signal or the power may then be transmitted or received between the communication module **190** and the external electronic device via the selected at least one antenna. According to an embodiment, other parts (e.g., radio frequency integrated circuit (RFIC)) than the radiator may be further formed as part of the antenna module **197**.

[0055] At least some of the above-described components may be coupled mutually and communicate signals (e.g., commands or data) therebetween via an inter-peripheral communication scheme (e.g., a bus, general purpose input and output (GPIO), serial peripheral interface (SPI), or mobile industry processor interface (MIPI)).

[0056] According to an embodiment, instructions or data may be transmitted or received between the electronic device **101** and the external electronic device **104** via the server **108** coupled with the second network **199**. Each of the electronic devices **102** and **104** may be a device of a same type as, or a different type, from the electronic device **101**. According to an embodiment, all or some of operations to be executed at the electronic device **101** may be executed at one or more of the external electronic devices **102**, **104**, or **108**. For example, if the electronic device **101** should perform a function or a service automatically, or in response to a request from a user or another device, the electronic device **101**, instead of, or in addition to, executing the function or the service, may request the one or more external electronic devices to perform at least part of the function or the service. The one or more external electronic devices receiving the request may perform the at least part of the function or the service requested, or an additional function or an additional service related to the request, and transfer an outcome of the performing to the electronic device **101**. The electronic device **101** may provide the outcome, with or without further processing of the outcome, as at least part of a reply to the request. To that end, a cloud computing, distributed computing, or client-server computing technology may be used, for example.

[0057] The electronic device according to various embodiments may be one of various types of

electronic devices. The electronic devices may include, for example, a portable communication device (e.g., a smart phone), a computer device, a portable multimedia device, a portable medical device, a camera, a wearable device, or a home appliance. According to an embodiment of the disclosure, the electronic devices are not limited to those described above.

[0058] It should be appreciated that various embodiments of the disclosure and the terms used therein are not intended to limit the technological features set forth herein to particular embodiments and include various changes, equivalents, or replacements for a corresponding embodiment. With regard to the description of the drawings, similar reference numerals may be used to refer to similar or related elements. As used herein, each of such phrases as “A or B,” “at least one of A and B,” “at least one of A or B,” “A, B, or C,” “at least one of A, B, and C,” and “at least one of A, B, or C,” may include all possible combinations of the items enumerated together in a corresponding one of the phrases. As used herein, such terms as “1st” and “2nd,” or “first” and “second” may be used to simply distinguish a corresponding component from another, and does not limit the components in other aspect (e.g., importance or order). It is to be understood that if an element (e.g., a first element) is referred to, with or without the term “operatively” or “communicatively”, as “coupled with,” “coupled to,” “connected with,” or “connected to” another element (e.g., a second element), it means that the element may be coupled with the other element directly (e.g., wiredly), wirelessly, or via a third element.

[0059] As used herein, the term “module” may include a unit implemented in hardware, software, or firmware, and may interchangeably be used with other terms, for example, “logic,” “logic block,” “part,” or “circuitry”. A module may be a single integral component, or a minimum unit or part thereof, adapted to perform one or more functions. For example, according to an embodiment, the module may be implemented in a form of an application-specific integrated circuit (ASIC).

[0060] Various embodiments as set forth herein may be implemented as software (e.g., the program **140**) including one or more instructions that are stored in a storage medium (e.g., internal memory **136** or external memory **138**) that is readable by a machine (e.g., the electronic device **101**). For example, a processor (e.g., the processor **120**) of the machine (e.g., the electronic device **101**) may invoke at least one of the one or more instructions stored in the storage medium, and execute it, with or without using one or more other components under the control of the processor. This allows the machine to be operated to perform at least one function according to the at least one instruction invoked. The one or more instructions may include a code generated by a compiler or a code executable by an interpreter. The machine-readable storage medium may be provided in the form of a non-transitory storage medium. Wherein, the term “non-transitory” simply means that the storage medium is a tangible device, and does not include a signal (e.g., an electromagnetic wave), but this term does not differentiate between where data is semi-permanently stored in the storage medium and where the data is temporarily stored in the storage medium.

[0061] According to an embodiment, a method according to various embodiments of the disclosure may be included and provided in a computer program product. The computer program products may be traded as commodities between sellers and buyers. The computer program product may be distributed in the form of a machine-readable storage medium (e.g., compact disc read only memory (CD-ROM)), or be distributed (e.g., downloaded or uploaded) online via an application store (e.g., Play Store™), or between two user devices (e.g., smart phones) directly. If distributed online, at least part of the computer program product may be temporarily generated or at least temporarily stored in the machine-readable storage medium, such as memory of the manufacturer's server, a server of the application store, or a relay server.

[0062] According to various embodiments, each component (e.g., a module or a program) of the above-described components may include a single entity or multiple entities. According to various embodiments, one or more of the above-described components may be omitted, or one or more other components may be added. Alternatively or additionally, a plurality of components (e.g., modules or programs) may be integrated into a single component. In such a case, according to

various embodiments, the integrated component may perform one or more functions of each of the plurality of components in the same or similar manner as they are performed by a corresponding one of the plurality of components before the integration. According to various embodiments, operations performed by the module, the program, or another component may be carried out sequentially, in parallel, repeatedly, or heuristically, or one or more of the operations may be executed in a different order or omitted, or one or more other operations may be added.

[0063] According to an embodiment of the disclosure, an electronic device **101** comprises a display (e.g., the display device **160**), a processor **120** functionally connected with the display, and a memory **130** functionally connected with the processor **120**. The memory **130** stores instructions configured to, when executed, enable the processor **120** to display a first image through the display, display one or more second images through the display while displaying the first image, select a third image from among the one or more second images, identify a value of at least one property of the third image, generate a filter for applying the value of the at least one property to an image, apply the value of the at least one property to the first image using the filter, display the first image, to which the value of the at least one property is applied, through the display, and store the filter in the memory **130**.

[0064] According to an embodiment of the disclosure, the instructions may be configured to enable the processor **120** to display the one or more second images through the display based on a user input to an object for generating the filter while displaying the first image.

[0065] According to an embodiment of the disclosure, the instructions may be configured to enable the processor **120** to identify a value of at least one property of an image portion when the third image may include a blank portion and the image portion.

[0066] According to an embodiment of the disclosure, the instructions may be configured to enable the processor **120** to designate a portion of the third image based on a user input, and identify a value of at least one property of the designated portion of the third image.

[0067] According to an embodiment of the disclosure, the instructions may be configured to enable the processor **120** to recognize at least one object in the third image, and identify a value of at least one property for an area including the at least one recognized object.

[0068] According to an embodiment of the disclosure, the instructions may be configured to enable the processor **120** to display, through the display, a user interface for adjusting a value of a first property among the at least one property after displaying the first image to which the value of the at least one property is applied, adjust the value of the first property based on a user input to the user interface, and generate the filter based on the adjusted value of the first property.

[0069] According to an embodiment of the disclosure, the instructions may be configured to enable the processor **120** to display one or more fourth images through the display after displaying the first image to which the value of the at least one property is applied, select at least one fifth image from among the one or more fourth images, and generate the filter based on a value of at least one property of the at least one fifth image.

[0070] According to an embodiment of the disclosure, the instructions may be configured to enable the processor **120** to display an image indicating the filter through the display after storing the filter in the memory **130**.

[0071] According to an embodiment of the disclosure, an electronic device **101** comprises a camera, a display, a processor **120** functionally connected with the camera and the display, and a memory **130** functionally connected with the processor **120**. The memory **130** stores instructions configured to, when executed, enable the processor **120** to display a first image input in real-time using the camera and images indicating one or more filters through the display, display one or more second images through the display, based on a user input for generating a filter while displaying the first image and the images, select a third image from among the one or more second images, identify a value of at least one property of the third image, generate a filter for applying the value of the at least one property to an image, apply the value of the at least one property to the first

image using the filter, display the first image, to which the value of the at least one property is applied, through the display, store the filter in the memory **130**, and display an image indicating the filter and the images through the display.

[0072] According to an embodiment of the disclosure, the instructions may be configured to enable the processor **120** to display the third image as the image indicating the filter.

[0073] According to an embodiment of the disclosure, the instructions may be configured to enable the processor **120** to display the first image, to which the value of the at least one property is applied, as the image indicating the filter.

[0074] According to an embodiment of the disclosure, the at least one property may include at least one of a color tone, brightness, saturation, image complexity, contrast, sharpness, shadow, or color temperature of an image.

[0075] FIG. **2** is a flowchart illustrating a method for providing a filter in an electronic device according to an embodiment of the disclosure.

[0076] Referring to FIG. **2**, in operation **201**, the processor **120** may display a first image through a display (e.g., the display device **160**).

[0077] The processor **120** may display the first image through the display using a camera application. For example, when receiving an input for executing the camera application from the user, the processor **120** may display a preview (or live view) image input in real time through the camera, as a first image, through the display. As another example, while displaying the preview image, the processor **120** may display an image captured based on a user input, as the first image, through the display.

[0078] The processor **120** may display the first image through the display using an image application (e.g., a Gallery application). For example, the processor **120** may display a plurality of images through the display by executing an image application. The processor **120** may display an image selected from among the plurality of images based on a user input, as a first image, through the display.

[0079] However, the method for displaying the first image is not limited to the above-described examples. For example, the processor **120** may display a screen captured based on an input for capturing the screen displayed (e.g., being currently displayed) through the display from the user, as the first image, through the display. The processor **120** may display an image (e.g., an image obtained from a webpage or an image registered (or posted) on a social media service server) obtained from the outside (e.g., an external server) using a web application or a social media service application, as the first image, through the display. The processor **120** may display an image received through the communication module **190** from an external electronic device or an image shared with the external electronic device, as the first image, through the display. The processor **120** may receive an image, which is obtained in real-time by an external electronic device (e.g., a camera or wearable device connected for communication with the electronic device) connected for communication with the electronic device, from the external electronic device and display the received image, as the first image, through the display.

[0080] The first image may be denoted an original image, target image, or input image to which an image property is to be applied using a filter to be generated.

[0081] In operation **203**, the processor **120** may display one or more second images through the display.

[0082] The processor **120** may display one or more second images through the display, based on a user input for generating a filter (hereinafter, a 'first user input').

[0083] The processor **120** may receive an input for generating a filter. For example, upon receiving a user input for editing the first image while displaying the first image, the processor **120** may display, through the display, an object for generating a filter, along with images individually indicating a plurality of pre-stored (or pre-generated) filters (or images individually corresponding to the plurality of filters). The processor **120** may receive a user input for the object for generating

a filter, as a first user input.

[0084] The processor **120** may display one or more second images, stored in the memory **130** (e.g., a Gallery application), through the display, based on the first user input. For example, the processor **120** may display one or more second images stored in the memory **130**, through the display, in a shrunk form (e.g., a thumbnail form), based on the first user input. However, methods for displaying one or more second images are not limited thereto. Hereinafter, one or more images to be displayed through the display based on the first user input are denoted as ‘one or more second images.’

[0085] Although in the above-described examples, the processor **120** display one or more second images stored in the memory **130** through the display based on the first user input, embodiments of the disclosure are not limited thereto. For example, the processor **120** may obtain one or more second images (e.g., images obtained from a webpage or images registered in a social media service server) through a web application or social media service application based on the first user input and display the one or more obtained second images through the display.

[0086] At least some (e.g., at least two) of the one or more second images displayed based on the first user input may be images. However, embodiments of the disclosure are not limited thereto; for example, at least some of the one or more second images displayed based on the first user input may be a videos (or moving images). At least some of the one or more second images displayed based on the first user input may be one frame (or frame image) included in a video (or included in a plurality of frames constituting a video).

[0087] In operation **205**, the processor **120** may select a third image from among one or more second images.

[0088] The processor **120** may receive a user input (hereinafter, referred to as a ‘second user input’) for selecting the third image from among one or more second images. For example, the processor **120** may receive a touch input for one or more images displayed in a shrunk form (e.g., a thumbnail form) from the user. In an embodiment, the processor **120** may select the third image based on the second user input. Hereinafter, an image selected from one or more second images based on the second user input is referred to as a ‘third image.’

[0089] In one embodiment of the disclosure, the third image may be an image. However, embodiments of the disclosure are not limited thereto, and in an embodiment, the third image may be a video (or moving image). For example, the third image may be one frame included in a video (or included in a plurality of frames constituting a video).

[0090] The third image may be referred to as a reference image that provides a property of the image (or from which a property of the image is to be extracted) to generate a filter.

[0091] In operation **207**, the processor **120** may identify (or extract) at least one property value of the third image.

[0092] At least one property (or at least one property of an image) may include a property that may be identified (or extracted) from the image (or image data). For example, at least one property may include at least one of the color tone (or hue), brightness, saturation (or chroma), image complexity, contrast, sharpness, shadow, or color temperature (or white balance) of the image. The at least one property may include at least a visual effect applied to the image. However, the at least one property is not limited to the above examples, and may include any property related to the image.

[0093] The processor **120** may identify at least one property value (e.g., a parameter related to at least one property) for at least a portion of the third image.

[0094] The processor **120** may designate a portion of the third image for which at least one property value is to be identified.

[0095] The processor **120** may designate (or identify) a portion of the third image for which at least one property value is to be identified, in response to a selection of the third image among one or more second images. For example, in response to the selection of the third image among one or more second images, if the third image includes a blank portion other than the image portion (or

image area), the processor **120** may designate the image portion other than the blank portion in the third image as the portion of the third image for which at least one property value is to be identified. In an embodiment, the processor **120** may crop the image portion except for the blank portion from the third image. In response to the selection of the third image among one or more second images, if the third image includes a text (e.g., a text included in the blank portion) other than the image portion, the processor **120** may designate the image portion other than the text-containing portion (or area) in the third image as the portion of the image for which at least one property value is to be identified. In an embodiment, the processor **120** may crop the image portion except for the text-containing portion from the third image.

[0096] The processor **120** may designate (or set) an image portion for which at least one property value is to be identified from the third image, based on a user input.

[0097] When a portion of the third image is designated from the third image, the processor **120** may identify at least one property value of the designated portion of the third image, based on the designated (or cropped) portion of the third image (or data for the designated portion of the third image).

[0098] A method in which the processor **120** designates an image portion for which at least one property value is to be identified from the third image based on a user input and/or in response to a selection of the third image and identifies at least one property value of the designated image portion is described below in detail with reference to FIG. 3.

[0099] According to an embodiment of the disclosure, the processor **120** may recognize an object (e.g., a figure or thing) included in the third image from the third image and identify at least one property of the recognized object (e.g., an area including at least a portion of the recognized object). For example, the processor **120** may recognize the object in the third image using a program (or algorithm) capable of recognizing the object in the image. The processor **120** may identify at least one property value of the area including at least a portion of the recognized object. A method for recognizing the object in the third image is described below in detail with reference to FIG. 4.

[0100] When the third image is a video, the processor **120** may identify at least one property of one frame of the third image. For example, when the third image is a video including a plurality of frames, the processor **120** may select one frame from among the plurality of frames, based on a user input (e.g., a user input for capturing one frame among the plurality of frames or a user input for temporarily stopping the playing video and then capturing a frame of the stopped video). The processor **120** may identify at least one property of the selected frame.

[0101] In operation **209**, the processor **120** may generate (or obtain) a filter for applying at least one property value to the image.

[0102] The processor **120** may generate a filter for applying at least one property value to the image based on at least one property value of the third image. For example, the processor **120** may generate a filter capable of applying at least one property value of the third image to the first image displayed through operation **201** and/or an image to be obtained after the first image.

[0103] The processor **120** may generate a filter for applying at least one property value to the image based on at least one property value of the image portion of the third image.

[0104] The processor **120** may generate a filter for applying at least one property value to the image based on at least one property value of the object included in the third image.

[0105] The processor **120** may generate a filter configured based on at least one property value of the third image (e.g., at least one property value identified based on data of the third image).

[0106] The processor **120** may generate a filter based on at least one property value of the third image and a plurality of designated (e.g., pre-generated) filters. For example, a plurality of different filters generated according to at least one property value related to the image may be stored in the memory **130**. The plurality of filters may be generated to differ from one another (or to correspond to at least one property value related to the image) according to at least one property value related

to the image, using artificial intelligence on the electronic device **101** or an external electronic device and may be stored in the memory **130**. However, methods for generating a plurality of filters are not limited thereto. The processor **120** may identify a filter corresponding to at least one property value of the third image among the plurality of filters stored in the memory **130**. The processor **120** may obtain the identified filter as a filter for applying at least one property value to the image.

[0107] When a filter is generated, the processor **120** may temporarily store the generated filter in the memory **130**. In one embodiment, the processor **120** may store the generated filter in a temporary memory (e.g., the volatile memory **132** or random access memory (RAM)).

[0108] In operation **211**, the processor **120** may apply at least one property value to the first image using the filter.

[0109] For example, the processor **120** may apply at least one property value of the third image to the first image displayed through operation **201** using the filter generated through operation **209**.

[0110] When the first image is a preview image input in real time through a camera, the processor **120** may apply at least one property value of the third image to the preview image input in real-time, using the generated filter.

[0111] When the first image is displayed through an image application, the processor **120** may apply at least one property value of the third image to the first image using the generated filter.

[0112] In operation **213**, the processor **120** may display the first image, to which at least one property value has been applied, through the display, using the filter.

[0113] When the first image is a preview image input in real-time through the camera, the processor **120** may display the preview image, to which at least one property value of the third image has been applied, through the display, using the generated filter.

[0114] When the first image is displayed through an image application, the processor **120** may display the first image, to which at least one property value of the third image has been applied, through the display, using the generated filter.

[0115] The processor **120** may display the third image, together with the first image to which at least one property value has been applied, through the display. For example, the processor **120** may display, through the display, the third image, together with the first image to which at least one property value has been applied, in a shrunk image form. The processor **120** may overlay and display the third image on the first image, to which at least one property value has been applied, through the display. The processor **120** may display the third image, along with the first image to which at least one property value has been applied, in an area independent (or separate) from the area where the at least one property value-applied first image is displayed, through the display.

[0116] After displaying the first image to which at least one property value has been applied, the processor **120** may adjust the values of one or more properties among the at least one property. A method for adjusting the values of one or more of the at least one property by the processor **120** is described below in detail with reference to FIGS. **5** and **6**.

[0117] The processor **120** may perform the operation of designating a portion of the third image, to which at least one property value is to be applied, based on a user input, after displaying the first image to which at least one property value has been applied. The processor **120** may perform the operation of generating a filter based on at least one property value of the designated portion of the third image, the operation of applying at least one property value of the designated portion of the third image to the first image using the generated filter, and the operation of displaying, through the display, the first image to which at least one property value of the designated portion of the third image has been applied. The operation of designating a portion of the third image to which at least one property value is to be applied based on the user input after displaying the first image to which at least one property value has been applied through the operation of displaying, through the display, the first image to which at least one property value of the designated portion of the third image has been applied are described below in detail with reference to FIG. **7**.

[0118] In operation **215**, the processor **120** may store the generated filter in the memory **130**.

[0119] The processor **120** may store the generated filter in the memory **130** based on a user input while displaying the first image to which at least one property value has been applied. For example, the processor **120** may, e.g., semi-permanently, store the generated filter in the memory **130** (e.g., the volatile memory **132**), based on a user input for an object for storing the generated filter while displaying the first image to which at least one property value has been applied.

[0120] Upon storing the generated filter in the memory **250**, the processor **120** may display the image corresponding to the generated filter through the display. For example, upon storing the generated filter in the memory **130**, the processor **120** may display an image indicating the generated filter through the display. Upon storing the generated filter in the memory **130**, the processor **120** may display, through the display, the generated filter along with images indicating a plurality of pre-stored filters.

[0121] The processor **120** may display the image corresponding to the generated filter through the display in various manners. For example, the processor **120** may display the third image, which is the basis of the generated filter, as the image corresponding to the generated filter, through the display. The processor **120** may display, through the display, the image corresponding to the first image to which at least one property value has been applied (e.g., a shrunk image of the first image to which at least one property value has been applied) using the generated filter, as the image corresponding to the generated filter. The processor **120** may display, through the display, an image to which at least one property value has been applied (e.g., any image that does not violate copyright) using the generated filter, as the image corresponding to the generated filter.

[0122] The processor **120** may display an image containing text indicating the generated filter and corresponding to the generated filter, through the display. For example, the processor **120** may display, through the display, an image corresponding to the generated filter and containing text indicating the name of the generated filter or text indicating the image property (e.g., at least one property of the third image) which is the basis of the generated filter. However, without limitations thereto, the processor **120** may display, through the display, an image that corresponds to the generated filter and does not include the text indicating the generated filter.

[0123] Although not shown in FIG. 2, the processor **120** may store the first image to which at least one property value has been applied in the memory **130**. For example, the processor may store the first image to which at least one property value has been applied in the memory **130** based on a user input.

[0124] Although not shown in FIG. 2, after the newly generated filter is stored, and when a filter function is executed on the image displayed using an image application or a newly obtained image using a camera application, the processor may display the filter, newly generated and stored, through the display. For example, after the newly generated filter is stored, and when a filter function is executed on the image displayed using an image application or a newly obtained image using a camera application, the processor may display, through the display, the filter, newly generated and stored, as a pre-stored filter (e.g., along with filters pre-stored before newly generating the filter).

[0125] FIG. 3 is a flowchart illustrating a method for identifying at least one property value of a third image, according to an embodiment of the disclosure.

[0126] Referring to FIG. 3, in operation **301**, according to an embodiment, the processor **120** may designate a portion of the third image.

[0127] The processor **120** may identify an image portion in the third image, in response to a user input for selecting the third image among one or more second images.

[0128] The processor **120** may designate (or identify) a portion of the third image for which at least one property value is to be identified, in response to a selection of the third image among one or more second images.

[0129] In response to the selection of the third image among one or more second images, if the

third image includes a blank portion other than the image portion (or image area), the processor **120** may designate the image portion other than the blank portion in the third image as the portion of the third image for which at least one property value is to be identified. For example, when the third image includes a blank portion in addition to the image portion (or image area), the processor **120** may identify (or detect) an edge (or outline) of the image portion in the third image. The processor **120** may designate (or identify) the image portion, other than the blank portion in the third image, as a portion for which at least one property is to be identified (or extracted) based on the identified edge of the image portion. The processor **120** may crop the designated portion (e.g., the image portion) from the third image. Although in the above example, a portion of the third image is designated by identifying an edge of the image portion in the third image, embodiments of the disclosure are not limited thereto. For example, other various methods may be adopted for designating the image portion, for which at least one property is to be identified, in the third image.

[0130] In response to the selection of the third image among one or more second images, if the third image includes a text (e.g., a text included in the blank portion) other than the image portion (or image area), the processor **120** may designate the portion (e.g., the image portion) other than the text-containing portion in the third image as the portion of the image for which at least one property value is to be identified. The processor **120** may crop the designated portion (e.g., the image portion), except for the text-containing portion, from the third image.

[0131] When the third image includes a blank portion or text other than the image portion (or image area), the processor **120** may automatically designate only the image portion, except for the blank portion or text in the third image, as the portion of the third image, for which at least one property value is to be identified, thereby generating a filter based only on the image portion in the third image.

[0132] The processor **120** may designate, set, or select the portion of the third image, for which at least one property value is to be identified, from the third image based on a user input.

[0133] When the third image is selected from among one or more second images, the processor **120** may display, through the display (e.g., the display device **160**), an object (or image object) (hereinafter, a 'first object') for designating the portion for which at least one property value is to be identified by a user input in the third image, along with the selected third image. The processor **120** may identify the portion of the third image designated (or set) by the position of the first object, based on a user input to the first object (e.g., a drag input to move the first object). The processor **120** may crop the identified portion of the third image. However, methods for designating the image portion for which at least one property value is to be identified from the third image based on a user input are not limited thereto.

[0134] The processor **120** may generate a filter based on the user's desired portion in the third image by designating the portion for which at least one property value is to be identified from the third image based on a user input. The processor **120** may generate a filter based on the image portion, except for the blank portion or text in the third image, by designating the portion for which at least one property value is to be identified from the third image based on a user input.

[0135] The operation of designating the portion for which at least one property value is to be identified from the third image based on a user input may be performed after performing the operation of designating (or identifying) the portion of the third image, for which at least one property value is to be identified, in response to the selection of the third image from among the one or more second images. For example, the processor **120** may designate a portion of the third image, for which at least one property value is to be display, in response to the selection of the third image from among the one or more second images and crop the designated portion of the third image. The processor **120** may display the cropped portion of the third image through the display. The processor **120** may redesignate a portion for which at least one property value is to be identified in the displayed portion of the third image, based on a user input to the displayed portion of the third image. However, without limitations thereto, the operation of designating the portion

for which at least one property value is to be identified from the third image based on a user input may be performed before performing the operation of designating (or identifying) the portion of the third image, for which at least one property value is to be identified, in response to the selection of the third image from among the one or more second images.

[0136] In operation **303**, the processor **120** may identify at least one property value for the designated portion of the third image.

[0137] When a portion of the third image is designated from the third image, the processor **120** may identify at least one property value (or parameter related to at least one property) of the designated portion of the third image, based on the designated (or cropped) portion of the third image (or data for the designated portion of the third image). For example, the processor **120** may identify the value of at least one property of the designated portion of the third image (e.g., at least one of the color tone (or hue), brightness, saturation, image complexity, contrast, sharpness, shadow, or color temperature (or white balance) of the image or a visual effect). However, at least one property identified by the processor is not limited to the above examples.

[0138] FIG. **4** is a flowchart illustrating a method for identifying at least one property value by recognizing an object recognized in a third image, according to an embodiment of the disclosure.

[0139] Referring to FIG. **4**, in operation **401**, according to an embodiment, the processor **120** may recognize an object (e.g., a figure object or thing object) in the third image. For example, the processor **120** may recognize the object in the third image using a program (or algorithm) capable of recognizing the object in the image.

[0140] The processor **120** may recognize the object in the third image in response to selection of the third image from among one or more second images. The processor **120** may recognize the object in the third image based on a user input (e.g., a touch input to a user interface (e.g., an icon) matched with a function for performing object recognition) while displaying the third image.

[0141] The processor **120** may recognize at least one object in the third image. The processor **120** may display, through the display (e.g., the display device **160**), an indication that at least one object recognized in the third image is recognized. When a plurality of objects are recognized in the third image, the processor **120** may display, through the display, an indicating that each of the plurality of recognized objects is recognized, for each of the plurality of objects. For example, when the plurality of objects are recognized in the third image, the processor **120** may display the outline of each of the plurality of objects for each of the plurality of objects, through the display. However, methods for displaying an indication that each of the plurality of recognized objects is recognized are not limited thereto.

[0142] When an object is recognized in the third image, the processor **120** may select the recognized object. When a plurality of objects are recognized in the third image, the processor **120** may select at least one object from among the plurality of objects based on a user input. For example, the processor **120** may select at least one object, to which the user's touch input has been made, from among the plurality of objects. However, methods for selecting an object recognized in the third image are not limited thereto.

[0143] In operation **403**, The processor **120** may identify at least one property value for the portion including the object in the third image.

[0144] The processor **120** may identify at least one property value of the portion including the recognized object (e.g., the portion corresponding to the recognized object) in the third image. For example, the processor **120** may identify the value of at least one property of the portion including the recognized object (e.g., at least one of the color tone (or hue), brightness, saturation, image complexity, contrast, sharpness, shadow, or color temperature (or white balance) of the image or a visual effect).

[0145] FIG. **5** is a flowchart illustrating a method for adjusting at least one property value based on a user interface, according to an embodiment of the disclosure.

[0146] The operations shown in FIG. **5** may be included in the operation of displaying the first

image to which at least one property value has been applied, as in operation **213** of FIG. 2, or may be operations performed after the performing the operation of displaying the first image to which at least one property value has been applied.

[0147] Referring to FIG. 5, in operation **501**, the processor **120** may select a first property from among at least one property.

[0148] While displaying the first image to which at least one property value has been applied, the processor **120** may select a property to be adjusted (hereinafter, a 'first property') from among at least one property, based on a user input. For example, upon receiving a user input for the third image while displaying the third image along with the first image to which at least one property value has been applied, the processor **120** may display, through the display (e.g., the display device **160**), at least one object corresponding to at least one property (or indicating at least one property). The processor **120** may select the first property corresponding to the user input-received object from among at least one object corresponding to at least one property.

[0149] In operation **503**, the processor **120** may display an interface for adjusting the value of the selected first property, through the display.

[0150] When the first property is selected, the processor **120** may display the interface for adjusting the value of the selected first property through the display.

[0151] To adjust the value of the first property, the processor **120** may display an interface including bar indicating the range of the first property value and an object, which may move according to the bar, through the display. The positions of the object on the bar may individually correspond to the values of the first property. To adjust the value of the first property, the processor **120** may display an interface including the field for inputting the value of the first property through the display. However, methods for displaying the interface for adjusting the value of the selected first property are not limited thereto.

[0152] In operation **505**, the processor **120** may generate a filter having an adjusted value of the first property (e.g., for which the adjusted value of the first property may be applied to the image), based on a user input to the interface.

[0153] The processor **120** may adjust the value of the first property based on a user input to the interface. For example, the processor **120** may adjust the value of the first property by moving the position of the object for adjusting the value of the first property according to the bar indicating the range of the value of the first property, based on a user input (e.g., a drag input to the object for adjusting the value of the first property). As another example, the processor **120** may adjust the value of the first property by inputting the value of the first property to the field for inputting the value of the first property based on a user input.

[0154] The processor **120** may generate a filter based on at least one property value including the adjusted value of the first property. For example, when at least one property which is the basis of a filter to be generated includes a first property and a second property and, of the values of the first property and the second property, the value of the first property is adjusted, the processor **120** may generate a filter for applying the adjusted value of the first property and the non-adjusted value of the second property to the image.

[0155] In operation **507**, the processor **120** may display, through the display, the first image using a filter generated based on at least one property value including the adjusted value of the first property.

[0156] Since the operation of displaying the first image using a filter generated based on at least one property value including the adjusted value of the first property as described above in connection with operation **507** is at least partially the same or similar to the operation of displaying the first image to which at least one property value has been applied using a filter as described above in connection with operation **213** of FIG. 2, no duplicate description is given below.

[0157] When the first image is a preview image input in real-time through the camera, the processor **120** may display the preview image, to which at least one property value including the

adjusted value of the first property of the third image has been applied, through the display, using the generated filter.

[0158] When the first image is displayed through an image application, the processor **120** may display the first image, to which at least one property value including the adjusted value of the first property of the third image has been applied, through the display, using the generated filter.

[0159] The processor **120** may display the third image, together with the first image to which at least one property value including the adjusted value of the first property has been applied, through the display.

[0160] The processor **120** may display the third image in which the value of the first property has been adjusted, together with the first image to which at least one property value including the adjusted value of the first property has been applied, through the display.

[0161] FIG. **6** is a flowchart illustrating a method for adjusting at least one property value based on a plurality of images, according to an embodiment of the disclosure.

[0162] Referring to FIG. **6**, the operations illustrated in FIG. **6** may be included in the operation of displaying the first image to which at least one property value has been applied, as in operation **213** of FIG. **2**, or may be operations performed after the performing the operation of displaying the first image to which at least one property value has been applied.

[0163] Referring to FIG. **6**, in operation **601**, the processor **120** may select a first property from among at least one property.

[0164] Since the operation of selecting the first property among at least one property by the processor **120** as in operation **601** is at least partially the same or similar to the operation of selecting the first property from among at least one property by the processor **120** as in operation **501** of FIG. **4**, no repetitive description thereof is given below.

[0165] In operation **603**, the processor **120** may display one or more fourth images through the display (e.g., the display device **160**).

[0166] In response to selection of the first property, the processor **120** may display one or more fourth images related to the first property stored in the memory **130** through the display.

[0167] When the processor **120** selects the first property from among at least one property, the processor **120** may display one or more fourth images, which have the value of the first property in a designated range from the value of the first property of the third image among one or more second images stored in the memory **130**, through the display. For example, when the processor **120** selects color tone (or hue) as the first property, the processor **120** may identify one or more fourth images having a color tone value in a designated range from the first value of the color tone of the third image among one or more second images stored in the memory **130** (hereinafter denoted as ‘one or more fourth images’). The processor **120** may display the one or more identified fourth images through the display.

[0168] When selecting the first property from among at least one property, the processor **120** may display one or more second images stored in the memory **130** through the display. The processor **120** may control the display to separately display one or more fourth images having the value of the first property, which belongs to a designated range from the value of the first property of the third image, among one or more second images and images having the value of the first property, which does not belong to the designated range from the value of the first property of the third image. For example, the processor **120** may control the display so that, among one or more second images, the one or more fourth images differ in shade from the images having the value of the first property which does not belong to the designated range from the value of the first property of the third image. However, methods for controlling the display to separately display one or more fourth images and the images having the value of the first property which does not belong to the designated range from the value of the first property of the third image among the one or more second images are not limited thereto. For example, the processor may display, through the display, only one or more fourth images which have a value belonging to the designated range from the

value of the first property of the third image among one or more second images.

[0169] The processor **120** may enable selection to be possible for one or more fourth images among one or more second images and not to be possible for the images having the value of the first property which does not belong to the designated range from the value of the first property of the third image. The processor **120** may enable inclusion of an object indicating that one or more fourth images are selectable among one or more second images, and exclusion of an object indicating that the images having the value of the first property which does not belong to the designated range from the value of the first property of the third image are selectable.

[0170] The processor **120** may enable selection to be rendered possible for one or more fourth images among one or more second images and not to be rendered possible for the images having the value of the first property which does not belong to the designated range from the value of the first property of the third image, thereby allowing the user to easily select at least some of one or more fourth images having the value of the first property, which are similar to the value of the first property of the third image (e.g., which fall within the designated range from the value of the first property of the third image). According to an embodiment, the one or more fourth images may include the third image.

[0171] In operation **605**, the processor **120** may select at least one fifth image from among one or more fourth images. For example, the processor **120** may select at least one fifth image from among the one or more fourth images based on a user input (e.g., a touch input to at least one fifth image among the one or more fourth images).

[0172] The processor **120** may select at least one fifth image, including the third image, from among one or more fourth images. The processor **120** may select at least one fifth image, which does not include the third image, from among one or more fourth images. For example, in operation **603**, upon displaying one or more fourth images, the third image may be in the state of being selected as default. The processor **120** may deselect the third image based on a user input and may select at least one image, except for the third image, as at least one fifth image, from among one or more fourth images.

[0173] In operation **607**, the processor **120** may generate a filter based on the value of the first property of at least one fifth image. For example, the processor **120** may identify the value of the first property of at least one fifth image. The processor **120** may determine (or calculate) the value of the first property which results from averaging the value of the first property of at least one fifth image. The processor **120** may generate a filter based on at least one property value including the determined (or calculated) value of the first property. Although in the above-described example, a filter is generated using the average of the value of the first property of at least one fifth image, embodiments of the disclosure are not limited thereto.

[0174] In operation **609**, the processor **120** may display, through the display, the first image using the filter generated based on the first property value of at least one fifth image.

[0175] Since the operation of displaying the first image using a filter generated based on at least one property value of at least one fifth image as in operation **609** is at least partially the same or similar to the operation of displaying the first image using a filter generated based on at least one property value including the adjusted value of the first property as in operation **507** of FIG. 5, no duplicate description is given below.

[0176] Although FIG. 6 illustrates an example operation for displaying one or more fourth images related to the first property, embodiments of the disclosure are not limited thereto. For example, the processor **120** may display, through the display, one or more second images stored in the memory **130** in response to selection of the first property in operation **601**. The processor **120** may select at least one fifth image from among one or more displayed second images based on a user input and generate a filter based on the first property value of at least one fifth image selected.

[0177] FIG. 7 is a flowchart illustrating a method for generating a filter based on a portion of a third image designated based on a user input, according to an embodiment of the disclosure.

[0178] Referring to FIG. 7, the operations illustrated in FIG. 7 may be included in the operation of displaying the first image to which at least one property value has been applied, as in operation 213 of FIG. 2, or may be operations performed after the performing the operation of displaying the first image to which at least one property value has been applied.

[0179] Referring to FIG. 7, in operation 701, the processor 120 may designate a portion of the third image based on a user input.

[0180] The processor 120 may display, through the display (e.g., the display device 160), a first object for designating a portion for which at least one property value is to be identified by a user input in the third image while displaying the third image along with the first image to which at least one property value has been applied. For example, the processor 120 may display, through the display, an enlarged third image and a first object in the enlarged third image, based on a user input for selecting the third image while displaying the third image along with the first image to which at least one property value has been applied. The processor 120 may identify the portion of the third image designated (or set) by the position of the first object, based on a user input to the first object (e.g., a drag input to move the first object). The processor 120 may crop the identified portion of the third image. However, methods for designating the image portion for which at least one property value is to be identified from the third image based on a user input are not limited thereto.

[0181] The processor 120 may generate a filter based on the user's desired portion in the third image by designating the portion for which at least one property value is to be identified from the third image based on a user input even after displaying the first image to which at least one property value has been applied.

[0182] In operation 703, the processor 120 may identify at least one property value for the designated portion of the third image.

[0183] In operation 705, the processor 120 may generate (or obtain) a filter for applying at least one property value to the image.

[0184] In operation 707, the processor 120 may apply at least one property value to the first image using the filter.

[0185] In operation 709, the processor 120 may display the first image, to which at least one property value has been applied, through the display, using the filter.

[0186] Since the embodiments of operations 703 to 709 are at least partially the same or similar to the embodiments of operations 207 to 213 of FIG. 2, no detailed description thereof is presented below.

[0187] FIG. 8 is a flowchart illustrating a method for providing a filter using a camera application according to an embodiment of the disclosure.

[0188] Referring to FIG. 8, in operation 801, the processor 120 may display a first image obtained through a camera through a display (e.g., the display device 160). For example, when receiving an input for executing the camera application from the user, the processor 120 may display a preview image input in real time through the camera, as a first image, through the display.

[0189] In operation 803, the processor 120 may display one or more second images through the display.

[0190] The processor 120 may receive a user input for displaying a filter for applying (or changing) at least one property to the first image while displaying the first image.

[0191] The processor 120 may display, through the display, an object for newly generating a filter along with images individually indicating (or corresponding to) a plurality of pre-stored filters based on a user input for displaying the filter for applying (or changing) at least one property to the first image.

[0192] The plurality of pre-stored filters may include at least one filter generated by the generator (or developer) of the camera application (hereinafter, 'at least one first filter') and at least one filter generated and stored by the user of the electronic device 101 (e.g., the filter stored through operation 215 of FIG. 2) (hereinafter, 'at least one second filter'). However, the at least one first

filter is not limited to filters generated by the generator (or developer) of the camera application but may rather include filters which have been downloaded by the electronic device after generated by a third party (or an external device).

[0193] The processor **120** may display at least one first filter through the display, based on a user input while displaying the first image. For example, the processor **120** may display, through the display, a shrunken image (e.g., image displayed in the form of a thumbnail) of the image where the property corresponding to each of at least one first filter has been applied to the first image.

[0194] The processor **120** may display, through the display, the object for generating a filter and at least one second filter based on a user input while displaying at least one first filter. For example, the processor **120** may display, through the display, the object for generating a filter along with a shrunken image of the third image (e.g., an image for which at least one property to be applied to the image has been identified (or extracted) upon generating at least one second filter) used to generate at least one second filter based on a user input while displaying at least one first filter.

[0195] Each of the images indicating the plurality of pre-stored filters may include text indicating the name of the filter or text indicating (or corresponding to) the image property (or the property value of the image) (e.g., at least one property of the third image) which is the basis of the filter.

[0196] However, methods for displaying the pre-stored filters by the processor **120** and methods for displaying the object for generating a filter are not limited thereto.

[0197] The processor **120** may display, through the display, one or more second images stored in the memory **130**, based on a user input to the object for generating a filter (e.g., a touch input to the object for generating a filter). For example, the processor **120** may display, through the display, one or more second images stored in the memory **130** and related to an image application or camera application, based on a user input to the object for generating a filter.

[0198] The processor **120** may display the second images in a shrunken form (e.g., a thumbnail form) through the display.

[0199] In operation **805**, the processor **120** may select a third image from among

[0200] one or more second images.

[0201] In operation **807**, the processor **120** may identify (or extract) at least one property value of the third image.

[0202] In operation **809**, the processor **120** may generate (or obtain) a filter for applying at least one property value to the image.

[0203] Since the embodiments of operations **805** to **809** are at least partially the same or similar to the embodiments of operations **205** to **209** of FIG. 2, no detailed description thereof is presented below.

[0204] In operation **811**, The processor **120** may apply at least one property value to the first image using the filter (e.g., the filter generated through operation **809**). The processor **120** may apply the at least one property value of the third image to the preview image input in real-time as the first image.

[0205] In operation **813**, The processor **120** may display the first image, to which at least one property value has been applied, through the display, using the filter.

[0206] The processor **120** may continuously display, through the display, the preview image to which at least one property value of the third image has been applied, for the preview image input in real-time as the first image.

[0207] The processor **120** may display the third image, together with the first image to which at least one property value has been applied, through the display. For example, the processor **120** may display, through the display, the third image, together with the first image to which at least one property value has been applied, in a shrunken image form. The processor **120** may overlay and display the third image on the first image, to which at least one property value has been applied, through the display. The processor **120** may display the third image, along with the first image to which at least one property value has been applied, in an area independent (or separate) from the

area where the at least one property value-applied first image is displayed, through the display. [0208] In operation **815**, according to an embodiment, the processor **120** may store the generated filter in the memory **130**. The processor **120** may store the generated filter in the memory **130** based on a user input while displaying the first image to which at least one property value has been applied. For example, the processor **120** may store the generated filter in the memory **130**, based on a user input for an object for storing the generated filter while displaying the first image to which at least one property value has been applied.

[0209] Upon storing the generated filter in the memory **250**, the processor **120** may display the image corresponding to the generated filter through the display. For example, upon storing the generated filter in the memory **130**, the processor **120** may display an image indicating the generated filter through the display. As another example, upon storing the generated filter in the memory **130**, the processor **120** may display, through the display, the generated filter along with the image indicating at least one second filter which has been displayed in operation **801**.

[0210] The processor **120** may display, through the display, a shrunken image of the third image (or a thumbnail image of the third image), as the image indicating the generated filter. However, embodiments of the disclosure are not limited thereto.

[0211] The processor **120** may display an image containing text indicating the generated filter and corresponding to the generated filter, through the display. For example, the processor **120** may display, through the display, an image corresponding to the generated filter and containing text indicating the name of the generated filter or text indicating the image property (e.g., at least one property of the third image) which is the basis of the generated filter.

[0212] Although not shown in FIG. **8**, the examples described above in connection with FIGS. **3** to **7** may apply likewise to the examples of the operations of FIG. **8**.

[0213] FIG. **9** is a flowchart illustrating a method for providing a filter using an image application according to an embodiment of the disclosure.

[0214] Referring to FIG. **9**, in operation **901**, the processor **120** may display a first image through a display (e.g., the display device **160**) using an image application (e.g., a Gallery application). For example, the processor **120** may display a plurality of images through the display by executing an image application. The processor **120** may display an image selected from among the plurality of images based on a user input, as a first image, through the display.

[0215] In operation **903**, the processor **120** may display one or more second images through the display.

[0216] The processor **120** may receive a user input for displaying a filter for applying (or changing) at least one property to the first image while displaying the first image.

[0217] The processor **120** may display, through the display, an object for newly generating a filter along with images individually indicating (or corresponding to) a plurality of pre-stored filters based on a user input for displaying the filter for applying (or changing) at least one property to the first image.

[0218] The plurality of pre-stored filters may include at least one filter generated by the generator (or developer) of the camera application (hereinafter, 'at least one first filter') and at least one filter generated and stored by the user of the electronic device **101** (e.g., the filter stored through operation **215** of FIG. **2**) (hereinafter, 'at least one second filter').

[0219] The processor **120** may display, through the display, at least one first filter, at least one second filter, and an object for generating a filter, based on a user input while displaying the first image. For example, the processor **120** may display, through the display, a shrunken image of the image for which the property corresponding to each of at least one first filter has been applied to the first image (e.g., an image displayed in the form of a thumbnail), a shrunken image of the third image used to generate at least one second filter (e.g., an image for which at least one property to be applied to the image has been identified), and an object for generating a filter.

[0220] Each of the images indicating the plurality of pre-stored filters may include text indicating

the name of the filter or text indicating the image property (or the property value of the image) (e.g., at least one property of the third image) which is the basis of the filter.

[0221] However, methods for displaying the pre-stored filters by the processor **120** and methods for displaying the object for generating a filter are not limited thereto.

[0222] The processor **120** may display, through the display, one or more second images stored in the memory **130**, based on a user input to the object for generating a filter (e.g., a touch input to the object for generating a filter). For example, the processor **120** may display, through the display, one or more second images stored in the memory **130** and related to an image application or camera application, based on a user input to the object for generating a filter.

[0223] The processor **120** may display the second images in a shrunken form (e.g., a thumbnail form) through the display.

[0224] In operation **905**, the processor **120** may select a third image from among one or more second images.

[0225] In operation **907**, the processor **120** may identify (or extract) at least one property value of the third image.

[0226] In operation **909**, the processor **120** may generate (or obtain) a filter for applying at least one property value to the image.

[0227] In operation **911**, the processor **120** may apply at least one property value to the first image using the filter (e.g., the filter generated through operation **909**).

[0228] In operation **913**, the processor **120** may display the first image, to which at least one property value has been applied, through the display, using the filter.

[0229] Since the embodiments of operations **905** to **913** are at least partially the same or similar to the embodiments of operations **205** to **213** of FIG. **2**, no detailed description thereof is presented below.

[0230] In operation **915**, according to an embodiment, the processor **120** may store the generated filter in the memory **130**. The processor **120** may store the generated filter in the memory **130** based on a user input while displaying the first image to which at least one property value has been applied. For example, the processor **120** may store the generated filter in the memory **130**, based on a user input for an object for storing the generated filter while displaying the first image to which at least one property value has been applied.

[0231] Upon storing the generated filter in the memory **250**, the processor **120** may display the image corresponding to the generated filter through the display. For example, upon storing the generated filter in the memory **130**, the processor **120** may display an image indicating the generated filter through the display. As another example, upon storing the generated filter in the memory **130**, the processor **120** may display, through the display, the generated filter along with images indicating pre-designated filters which have been displayed in operation **901**.

[0232] The processor **120** may display, through the display, a shrunken image of the first image, for which at least one property value has been applied, (or a thumbnail image of the first image), as the image indicating the generated filter. However, embodiments of the disclosure are not limited thereto.

[0233] The processor **120** may display an image containing text indicating the generated filter and corresponding to the generated filter, through the display. For example, the processor **120** may display, through the display, an image corresponding to the generated filter and containing text indicating the name of the generated filter or text indicating the image property (e.g., at least one property of the third image) which is the basis of the generated filter.

[0234] Although not shown in FIG. **9**, the examples described above in connection with FIGS. **3** to **7** may apply likewise to the examples of the operations of FIG. **9**.

[0235] FIGS. **10A** and **10B** are views illustrating a method for providing a filter using a camera application according to various embodiments of the disclosure.

[0236] Referring to FIGS. **10A** and **10B**, reference number **1001** may denote a screen including a

first image **1010** and images corresponding to pre-stored filters (e.g., at least one first filter).

[0237] The processor **120** may display, through the display (e.g., the display device **160**), the first image **1010**, as a preview image input in real-time, through a camera application. The processor **120** may receive a user input for displaying the filter for applying at least one property to the first image **1010**. As shown in **1001**, the processor **120** may display, through the display, the first image **1010**, an object **1011** for capturing (or recording) the first image **1010**, an object **1013** for displaying at least one first filter generated by the generator (or developer) of the camera application, an object **1014** for displaying an object for newly generating a filter and at least one second filter generated and stored by the user of the electronic device **101**, and images (e.g., an image **1015-2** and an image **1015-3**) indicating at least one first filter.

[0238] The processor **120** may display, through the display, the shrunken images (e.g., images displayed in the form of a thumbnail) (e.g., the image **1015-2** and the image **1015-3**) of the image for which the property corresponding to each of at least one first filter has been applied for the first image **1010**. The processor **120** may display, through the display, the image **1015-2** and the image **1015-3** including text (e.g., “warm” and “calm”) indicating at least one property applied to each of the images **1015-2** and the image **1015-3**. The processor **120** may display, through the display, the shrunken image **1015-1** of the first image, to which at least one property has not been applied for the first image **1010**. According to an embodiment, the processor **120** may display, through the display, the shrunken image **1015-1** of the first image **1010**, so that the shrunken image **1015-1** of the first image **1010** includes text (e.g., “original copy”) indicating that at least one property has not been applied.

[0239] Reference number **1003** may denote a screen including a first image **1020**, images corresponding to pre-stored filters (e.g., at least one second filter), and an object **1021-1** for newly generating a filter. Reference number **1003** may denote a screen that is displayed through the display when a user input (e.g., a touch input) to the object **1014** in **1001** is received.

[0240] As shown in **1003**, the processor **120** may display, through the display, the first image **1020**, images (e.g., the image **1021-3** and the image **1021-4**) indicating at least one second filter generated and stored by the user of the electronic device **101**, an object **1021-1** for newly generating a filter, a shrunken image **1021-2** of the first image, the object **1011**, the object **1013**, and the object **1014**.

[0241] The processor **120** may display, through the display, the image indicating at least one second filter, in the form of a shrunken form of the image for which at least one property has been identified (or extracted) upon generating at least one second filter. The processor **120** may display, through the display, the image **1021-3** and the image **1021-4** including the names (e.g., “filter1” and “filter2”) indicating the image **1021-3** and the image **1021-4**, respectively. The processor **120** may display, through the display, the shrunken image **1021-2** of the first image **1020**, so that the shrunken image **1021-1** of the first image **1020** includes text (e.g., “original copy”) indicating that at least one property has not been applied.

[0242] Reference number **1005** may denote a screen including one or more second images **1030**. Reference number **1005** may denote a screen that is displayed through the display when a user input (e.g., a touch input) to the object **1021-1** in **1003** is received.

[0243] The processor **120** may display one or more second images **1030** stored in the memory **130**, through the display, in a shrunken form (e.g., a thumbnail form). For example, the processor **120** may display, through the display, one or more second images **1030** stored in the memory **130** and related to an image application or camera application, based on a user input to the object **1021-1** for generating a filter.

[0244] Reference number **1007** may denote a screen displayed when the image **1031** is selected from among one or more second images **1030**.

[0245] According to an embodiment, the processor **120** may select the image **1031**, as the third image, based on a user input (e.g., a touch input to the image **1031**) among one or more second

images **1030**. The processor **120** may identify at least one property value of the selected image **1031**. The processor **120** may generate a filter based on at least one property value of the image **1031**.

[0246] The processor **120** may display, through the display, the first image **1040** to which at least one property value of the selected image **1031** has been applied using the generated filter as shown in **1007**.

[0247] The processor **120** may display, through the display, the selected image **1031** along with the first image **1040**. For example, the processor **120** may overlay and display the image **1031**, as the third image, on the first image **1040**, to which at least one property value has been applied, through the display. However, methods for displaying the third image are not limited to the above-described examples.

[0248] The processor **120** may display, through the display, the text (e.g., “filter7”) **1041** indicating the name of the newly generated filter in the first image **1040**.

[0249] The processor **120** may display, through the display, an object **1043** for storing the generated filter in the memory **130** and an object **1045** for deleting the generated filter (e.g., a filter generated and then temporarily stored in the memory **130**).

[0250] Reference number **1009** may denote a screen that is displayed through the display after the generated filter is stored. According to an embodiment, reference number **1009** may denote a screen that is displayed through the display when a user input (e.g., a touch input) to the object **1043** in **1007** is received.

[0251] The processor **120** may display, through the display, the first image **1050** to which at least one property value of the image **1031** has been applied using the generated filter and an image **1051** indicating the generated filter. For example, the processor **120** may display, through the display, the first image **1050** to which at least one property value of the image **1031** has been applied using the generated filter, an image indicating at least one pre-stored second filter (e.g., at least one second filter generated by the user of the electronic device **101**) (e.g., the image **1021-5** and the image **1021-6**), and an image **1051** indicating the generated filter.

[0252] The processor **120** may also display, through the display, an interface **1053** for adjusting the degree (or strength) of at least one property value applied by the filter (e.g., the generated filter or a filter selected from among pre-stored filters) for the image **1050**. The processor **120** may adjust the degree of at least one property value applied to the first image **1050**, being displayed through the display, using the generated filter, according to the position of the object **1053-1** included in the interface **1053**. For example, when the object **1053-1** is positioned at the leftmost side on the bar of the interface **1053** based on a user input to the object **1053-1**, the processor **120** may display an image (e.g., the image **1010** or original image), to which at least one property value has not been applied by the generated filter. When the object **1053-1** is positioned at the rightmost side on the bar of the interface **1053** based on a user input to the object **1053-1**, the processor **120** may display an image (e.g., the image **1050**), to which at least one property value has been fully applied by the generated filter.

[0253] FIGS. **11A** and **11B** are views illustrating a method for providing a filter using an image application according to various embodiments of the disclosure.

[0254] Referring to FIGS. **11A** and **11B**, reference number **1101** may denote a screen including a first image **1110**, an object **1111** for generating a filter, and images (e.g., an image **1112-2**, an image **1112-3**, and an image **1112-4**) corresponding to pre-stored filters (e.g., at least one first filter and at least one second filter).

[0255] When an image application is executed, the processor **120** may display the first image **1110** selected by the user through a display (e.g., the display device **160**). The processor **120** may display, through the display, a screen as shown in **1101**, based on an input (e.g., an input for using a filter on the selected first image **1110**) for editing the selected first image **1110**.

[0256] As shown in **1101**, the processor **120** may display, through the display, the first image **1110**,

images (e.g., the image **1112-2**, the image **1112-3**, and the image **1112-4**) indicating a plurality of pre-stored filters, and an object **1111** for generating a filter.

[0257] The processor **120** may display, through the display, the shrunk images (e.g., the image **1112-2**, the image **1112-3**, and the image **1112-4**) of the image for which the properties individually corresponding to the plurality of filters have been applied for the first image **1110**. The processor **120** may display, through the display, the image **1112-2**, the image **1112-3**, and the image **1112-4** including text (e.g., “automatic,” “warm,” and “calm”) indicating at least one property applied to each of the images **1112-2**, the image **1112-3**, and the image **1112-4**. The processor **120** may display, through the display, the shrunk image **1112-1** of the first image **1110**, to which at least one property has not been applied for the first image. The processor **120** may display, through the display, the shrunk image **1015-1** of the first image **1110**, so that the shrunk image **1112-1** of the first image **1110** includes text (e.g., “original copy”) indicating that at least one property has not been applied. The processor **120** may also display, through the display, an interface **1113** for adjusting the degree (or strength) of at least one property value applied by a filter selected from among a plurality of filters for the image **1110**. When no filter is selected, the processor **120** may display, through the display, an interface **1113** to indicate that the interface **1113** is unable to receive a user input (e.g., the interface **1113** deactivated). For example, when no filter is selected, the processor **120** may control the display to display the interface **1113** in a first color (e.g., black) and, when a filter is selected, control the display to display the interface **1113** in a second color (e.g., yellow) to indicate that the interface **1113** is able to receive user inputs.

[0258] Reference number **1103** may denote a screen including one or more second images **1120**. Reference number **1103** may denote a screen that is displayed through the display when a user input (e.g., a touch input) to the object **1111** in **1101** is received.

[0259] The processor **120** may display one or more second images **1120** stored in the memory **130**, through the display, in a shrunk form (e.g., a thumbnail form). For example, the processor **120** may display, through the display, one or more second images **1120** stored in the memory **130** and related to an image application, based on a user input to the object **1111** for generating a filter.

[0260] Reference number **1105** may denote a screen displayed when the image **1121** is selected from among one or more second images **1120**.

[0261] The processor **120** may select the image **1121**, as the third image, based on a user input (e.g., a touch input to the image **1121**) among one or more second images **1120**. The processor **120** may identify at least one property value of the selected image **1121**. The processor **120** may generate a filter based on at least one property value of the image **1121**.

[0262] The processor **120** may display, through the display, the first image **1130** to which at least one property value of the selected image **1121** has been applied using the generated filter as shown in **1105**.

[0263] The processor **120** may display, through the display, the selected image **1121** along with the first image **1130**. For example, the processor **120** may display, through the display, the image **1121**, as the third image, in an area independent from the area of the first image **1130** to which at least one property value has been applied, along with the first image **1130** to which at least one property value has been applied. However, methods for displaying the third image are not limited to the above-described examples.

[0264] The processor **120** may display, through the display, the text (e.g., “filter **1**”) **1125** indicating the name of the newly generated filter in the first image **1130**.

[0265] The processor **120** may display, through the display, an object **1131-3** for storing the generated filter in the memory **130** and an object **1131-1** for deleting the generated filter (e.g., a filter generated and then temporarily stored in the memory **130**).

[0266] Reference number **1107** may denote a screen that is displayed through the display after the generated filter is stored. According to an embodiment, reference number **1007** may denote a screen that is displayed through the display when a user input (e.g., a touch input) to the object

1131-3 in **1005** is received.

[0267] According to an embodiment, the processor **120** may display, through the display, the first image **1140** to which at least one property value of the image **1121** has been applied using the generated filter and an image **1141** indicating the generated filter. For example, the processor **120** may display, through the display, a first image **1140** to which at least one property value of the image **1121** has been applied using the generated filter, a shrunk image **1112-1** of the first image **1110**, a plurality of pre-stored filters (e.g., an image **1112-2** and an image **1112-3**), and an image **1141** indicating the generated filter.

[0268] The processor **120** may display, through the display, an interface **1113** for adjusting the degree (or strength) of at least one property value applied by the generated filter to the image **1140** to indicate that the interface **1113** is able to receive user inputs (or activated). The processor **120** may adjust the degree of at least one property value applied to the first image **1040**, being displayed through the display, using the generated filter, according to the position of the object **1113-1** included in the interface **1113**.

[0269] FIG. **12** is a view illustrating an example method for identifying at least one property value of a third image, according to an embodiment of the disclosure.

[0270] Referring to FIG. **12**, it is a view related to FIGS. **3** and **7**, illustrating example methods for generating a filter based on a portion of a third image when the portion of the third image is designated.

[0271] Referring to FIG. **12**, reference number **1201** may denote a screen including one or more second images. For example, the processor **120** may execute a camera application or an image application and display the first image through the display (e.g., the display device **160**). The processor **120** may display, through the display, one or more second images based on a user input to an object (e.g., an object **1021-1** or an object **1111**) for generating a filter.

[0272] The processor **120** may receive a user input for selecting the image **1211**, as the third image, from among one or more second images **1210**. The image **1211** may include a blank portion **1211-1** including text **1211-2** in addition to an image portion **1211-3**.

[0273] In response to selection of the image **1211**, the processor **120** may designate the image portion **1211-3**, except for the blank portion **1211-1** in the image **1211**, as the portion of the third image for which at least one property value is to be identified. According to an embodiment, the processor **120** may crop only the image portion **1211-3** in the image **1211**.

[0274] The processor **120** may identify at least one property value for the image portion **1211-3** in the image **1211**. The processor **120** may generate a filter based on at least one property value for the image portion **1211-3** in the image **1211**.

[0275] The processor **120** may display, through the display, the first image **1220** to which at least one property value has been applied for the image portion **1211-3**, using the generated filter. The processor **120** may display, through the display, text **1221** (e.g., “filter1”) indicating the name of the generated filter on the first image **1220**.

[0276] The processor **120** may display, through the display, the image portion **1211-3** along with the first image **1220**. According to an embodiment, the processor **120** may further display, through the display, an object **1223-3** for storing the generated filter in the memory **130** and an object **1223-1** for deleting the generated filter (e.g., a filter generated and then temporarily stored in the memory **130**).

[0277] Reference number **1205** may denote a screen for designating a portion of the third image (e.g., the image portion **1211-3**) based on a user input. According to an embodiment, reference number **1205** may denote a screen displayed when a user input (e.g., a touch input) to the image portion **1211-3** is received in **1203**. However, methods for displaying a screen in **1205** are not limited thereto.

[0278] The processor **120** may display, through the display, an object **1231** for designating a portion for which at least one property value is to be display by a user input in the image portion

1211-3 while displaying the image portion **1211-3** along with the first image **1220** to which at least one property value has been applied. For example, the processor **120** may display, through the display, an enlarged image portion **1230** and an object **1231** in the enlarged image portion **1230** (or the outline of the enlarged image portion **1230**), based on a user input for selecting the image portion **1211-3** while displaying the image portion **1211-3** along with the first image **1220** to which at least one property value has been applied. The processor **120** may identify a portion of the image portion **1211-3** designated (or set) by the position of the object **1231**, based on a user input to the object **1231** (e.g., a drag input for moving the object **1231**). The processor **120** may crop the identified portion of the image portion **1211-3**. However, methods for designating the image portion for which at least one property is to be identified from the image portion **1211-3** based on a user input are not limited thereto.

[0279] The processor **120** may designate an image portion for which at least one property value is to be identified from the image portion **1211-3** and generate a filter based on the designated image portion.

[0280] Although not shown in FIG. **12**, when the image **1211** is selected from among one or more second images **1210**, the processor **120** may designate an image portion, for which at least one property value is to be identified, based on a user input in the image **1211**. For example, when the image **1211** is selected from among one or more second images **1210**, the processor **120** may display, through the display, an object for designating the portion, for which at least one property value is to be identified by a user input in the image **1211**, along with the image **1211** and the first image **1220**. The processor **120** may designate the portion, for which at least one property value is to be identified in the image **1211**, based on a user input to the displayed object. The processor **120** may identify at least one property value of the designated portion and generate a filter based on the at least one identified property value.

[0281] Although not shown in FIG. **12**, when the image **1211** is selected from among one or more second images **1210**, the processor **120** may identify the image portion **1211-3** in the image **1211** and designate the image portion for which at least one property value is to be identified based on a user input in the image portion **1211-3**. The processor **120** may identify at least one property value of the designated portion and generate a filter based on the at least one identified property value.

[0282] FIG. **13** is a view illustrating an example method for identifying at least one property value by recognizing an object recognized in a third image, according to an embodiment of the disclosure.

[0283] FIG. **13** may be a view related to FIG. **4**, illustrating example methods for identifying at least one property value by recognizing an object recognized in the third image.

[0284] Referring to FIG. **13**, reference number **1301** may denote a screen including one or more second images. For example, the processor **120** may execute a camera application or an image application and display the first image **1310** through the display (e.g., the display device **160**). The processor **120** may display, through the display, one or more second images based on a user input to an object (e.g., an object **1021-1** or an object **1111**) for generating a filter.

[0285] Reference number **1303** may denote a screen displayed when the image **1311** is selected in **1301**.

[0286] The processor **120** may recognize an object (e.g., a figure object or thing object) in the image **1311**. For example, the processor **120** may recognize a figure's face as an object in the image **1311** using a program (or algorithm) capable of recognizing objects in an image.

[0287] The processor **120** may recognize an object in the image **1311** based on a user input (e.g., a touch input to a user interface matched with a function for performing object recognition) while displaying the first image **1320** and the image **1311**. The processor **120** may display, through the display, an indication **1321** that at least one object recognized in the image **1311** is recognized. When a plurality of objects are recognized in the image **1311**, the processor **120** may display, through the display, an indication that each of the plurality of recognized objects is recognized, for

each of the plurality of objects. For example, when the plurality of objects are recognized in the third image, the processor **120** may display the outline of each of the plurality of objects for each of the plurality of objects, through the display. However, methods for displaying an indication that each of the plurality of recognized objects is recognized are not limited thereto.

[0288] When an object is recognized in the image **1311**, the processor **120** may select the recognized object. When a plurality of objects are recognized in the image **1311**, the processor **120** may select at least one object from among the plurality of objects based on a user input. For example, the processor **120** may select at least one object, to which the user's touch input has been made, from among the plurality of objects. However, methods for selecting an object recognized in the image **1311** are not limited thereto.

[0289] According to an embodiment, the processor **120** may identify at least one property value for the portion including an object in the third image and generate a filter based on the at least one identified property value.

[0290] FIG. **14** is a view illustrating an example method for adjusting at least one property value based on a user interface, according to an embodiment of the disclosure.

[0291] Referring to FIG. **14**, while displaying the first image **1410** to which at least one property value has been applied, the processor **120** may select a property to be adjusted from among at least one property, based on a user input. For example, upon receiving a user input to the third image **1415** while displaying the third image **1415** along with the first image **1410** to which at least one property value has been applied, the processor **120** may display, through the display (e.g., the display device **160**), an interface **1411** including at least one object (e.g., an object **1412-1**, an object **1412-2**, an object **1412-3**, and an object **1412-4**) corresponding to (or indicating) at least one property, as shown in **1401**. The processor **120** may select an object (e.g., the object **1412-1**) indicating a first property (e.g., color tone or hue) in the interface **1411** based on a user input.

[0292] The processor **120** may further display, through the display, an object **1413-3** for storing the generated filter in the memory **130** an object **1413-1** for deleting the generated filter (e.g., a filter generated and then temporarily stored in the memory **130**), along with the first image **1410**, the third image **1415**, an object **1417** identifying the generated filter, and the interface **1411**.

[0293] When an object (e.g., the object **1421-1**) is selected as shown in **1403**, the processor **120** may display, through the display, an interface **1421** for adjusting the value of the selected first property. For example, to adjust the value of the first property, the processor **120** may display, through the display, an interface **1421** including a bar **1423-1** indicating a range of the first property value and an object **1423-2** that is movable along the bar **1423-1**. According to an embodiment, the positions of the object **1423-2** on the bar **1423-1** may individually correspond to the values of the first property. To adjust the value of the first property, the processor **120** may display an interface **1421** including the field **1423-3** for inputting the value of the first property through the display. However, methods for displaying the interface for adjusting the value of the selected first property are not limited thereto.

[0294] The processor **120** may adjust the first property value of the third image **1415** based on the value of the first property corresponding to the position of the object **1423-2** on the bar **1423-1**, based on a user input.

[0295] The processor **120** may generate a filter that may apply the adjusted value of the first property based on a user input to the interface **1421**. The processor **120** may display, through the display, the first image **1420** to which at least one property value has been applied including the adjusted value of the first property, using the generated filter.

[0296] FIG. **15** is a view illustrating an example method for adjusting at least one property value based on a plurality of images, according to an embodiment of the disclosure.

[0297] Referring to FIG. **15**, while displaying the first image **1510** to which at least one property value has been applied, the processor **120** may select a property to be adjusted from among at least one property, based on a user input. For example, upon receiving a user input to the third image

1515 while displaying the third image **1515** along with the first image **1510** to which at least one property value has been applied, the processor **120** may display, through the display, an interface **1511** including at least one object (e.g., an object **1512-1**, an object **1512-2**, an object **1512-3**, and an object **1512-4**) corresponding to (or indicating) at least one property, as shown in **1501**. The processor **120** may select an object (e.g., the object **1512-1**) indicating a first property (e.g., color tone or hue) in the interface **1511** based on a user input.

[0298] The processor **120** may further display, through the display, an object **1513-3** for storing the generated filter in the memory **130** and an object **1513-1** for deleting the generated filter (e.g., a filter generated and then temporarily stored in the memory **130**), along with the first image **1510**, the third image **1515**, and the interface **1511**.

[0299] In response to selection of the object **1521-1**, the processor **120** may display one or more fourth images related to the first property stored in the memory **130** through the display.

[0300] When the processor **120** selects the first property from among at least one property, the processor **120** may display one or more fourth images, which have the value of the first property in a designated range from the value of the first property of the image **1515** among one or more second images stored in the memory **130**, through the display. For example, when the processor **120** selects color tone (or hue) as the first property, the processor **120** may identify one or more fourth images having a color tone value in a designated range from the first value of the color tone of the image **1515** among one or more second images stored in the memory **130**. The processor **120** may display the one or more identified fourth images through the display.

[0301] As shown in **1503**, the processor **120** may control the display to separately display one or more fourth images having the value of the first property, which belongs to a designated range from the value of the first property of the image **1515**, among one or more second images **1520** and images having the value of the first property, which does not belong to the designated range from the value of the first property of the image **1515**. For example, the processor **120** may control the display so that, among one or more second images **1520**, the one or more fourth images differ in shade from the images having the value of the first property which does not belong to the designated range from the value of the first property of the image **1515**. However, methods for controlling the display to separately display one or more fourth images and the images having the value of the first property which does not belong to the designated range from the value of the first property of the image **1515** among the one or more second images **1520** are not limited thereto.

[0302] The processor **120** may enable selection to be possible for one or more fourth images among one or more second images **1520** and not to be possible for the images having the value of the first property which does not belong to the designated range from the value of the first property of the image **1515**. According to an embodiment, the processor **120** may enable inclusion of an object indicating that one or more fourth images are selectable among one or more second images **1520**, and exclusion of an object indicating that the images having the value of the first property which does not belong to the designated range from the value of the first property of the image **1515** are selectable.

[0303] The processor **120** may select at least one fifth image (e.g., an image **1521**, an image **1522**, and an image **1515**) from among one or more fourth images. For example, the processor **120** may select at least one fifth image from among the one or more fourth images based on a user input (e.g., a touch input to at least one fifth image among the one or more fourth images).

[0304] According to an embodiment, the processor **120** may select at least one fifth image, including the image **1515**, from among one or more fourth images. According to an embodiment, the processor **120** may select at least one fifth image, which does not include the image **1515**, from among one or more fourth images. For example, upon displaying one or more fourth images, the image **1515**, as the third image, may be in the state of being selected as default. The processor **120** may deselect the image **1515** based on a user input and may select at least one image, except for the image **1515**, as at least one fifth image, from among one or more fourth images. The processor **120**

may display, through the display, a selection of the image **1521**, image **1522**, and image **1515**, as one fifth image, through the object **1521-1**, object **1522-1**, and object **1515-1**.

[0305] The processor **120** may generate a filter based on the value of the first property of at least one fifth image. For example, the processor **120** may identify the value of the first property of at least one fifth image. The processor **120** may determine (or calculate) the value of the first property which results from averaging the value of the first property of at least one fifth image. The processor **120** may generate a filter based on at least one property value including the determined value of the first property. Although in the above-described example, a filter is generated using the average of the value of the first property of at least one fifth image, embodiments of the disclosure are not limited thereto.

[0306] As shown in **1505**, the processor **120** may display, through the display, the first image **1530** using a filter generated based on the value of the first property determined based on at least one fifth image (e.g., the image **1521**, the image **1522**, and the image **1515**).

[0307] A method according to an embodiment of the disclosure comprises displaying a first image through a display (e.g., the display device **160**) of an electronic device **101**, displaying one or more second images through the display while displaying the first image, selecting a third image from among the one or more second images, identifying a value of at least one property of the third image, generating a filter for applying the value of the at least one property to an image, applying the value of the at least one property to the first image using the filter, displaying the first image, to which the value of the at least one property is applied, through the display, and storing the filter in a memory **130** of the electronic device **101**.

[0308] According to an embodiment of the disclosure, displaying the one or more second images may include displaying the one or more second images through the display based on a user input to an object for generating the filter while displaying the first image.

[0309] According to an embodiment of the disclosure, identifying the value of the at least one property of the third image may include identifying a value of at least one property of an image portion when the third image includes a blank portion and the image portion.

[0310] According to an embodiment of the disclosure, identifying the value of the at least one property of the third image may include designating a portion of the third image based on a user input, and identifying a value of at least one property of the designated portion of the third image.

[0311] According to an embodiment of the disclosure, identifying the value of the at least one property of the third image may include recognizing at least one object in the third image, and identifying a value of at least one property for an area including the at least one recognized object.

[0312] According to an embodiment of the disclosure, the method may comprise displaying, through the display, a user interface for adjusting a value of a first property among the at least one property after displaying the first image to which the value of the at least one property is applied, adjusting the value of the first property based on a user input to the user interface, and generating the filter based on the adjusted value of the first property.

[0313] According to an embodiment of the disclosure, the method may comprise displaying one or more fourth images through the display after displaying the first image to which the value of the at least one property is applied, selecting at least one fifth image from among the one or more fourth images, and generating the filter based on a value of at least one property of the at least one fifth image.

[0314] According to an embodiment of the disclosure, the method may comprise displaying an image indicating the filter through the display after storing the filter in the memory **130**.

[0315] The structure of the data used in embodiments of the disclosure may be recorded in a computer-readable recording medium via various means. The computer-readable recording medium includes a storage medium, such as a magnetic storage medium (e.g., a ROM, a floppy disc, or a hard disc) or an optical reading medium (e.g., a CD-ROM or a digital versatile disc (DVD)).

[0316] While the disclosure has been shown and described with reference to various embodiments

thereof, it will be understood by those skilled in the art that various changes in form and details may be made therein without departing from the spirit and scope of the disclosure as defined by the appended claims and their equivalents.

Claims

1. An electronic device comprising: a display; at least one processor comprising processing circuitry; and memory storing instructions that, when executed by the at least one processor individually or collectively, cause the electronic device to: display, via the display, a first image, receive a first user input for editing the first image while the first image is displayed via the display, based at least in part on the first user input, display, via the display and concurrently with the first image, a first user interface (UI) object for generating a new filter, based at least in part on a second user input with respect to the first UI object, display, via the display, a plurality of images previously stored in the memory, based at least in part on a third user input with respect to a second image among the plurality of images, generate, as the new filter, a first filter corresponding to the second image, based on the generating of the first filter, display, via the display, a third image and a second UI object, the third image corresponding to the first image, the second UI object corresponding to the first filter, the third image being generated by applying the first filter to the first image, the second UI object including a thumbnail image indicative of the second image, and store the first filter in the memory.
2. The electronic device of claim 1, wherein the second UI object further includes a text indicative of the first filter.
3. The electronic device of claim 1, wherein the instructions, when executed by the at least one processor individually or collectively, cause the electronic device to: based at least in part on the first user input, display, via the display and concurrently with the first image and the first UI object, a third UI object corresponding to a second filter previously stored in the memory.
4. The electronic device of claim 3, wherein the third UI object includes a thumbnail image and a text, and wherein each of the thumbnail image and the text is indicative of the second filter.
5. The electronic device of claim 3, wherein the instructions, when executed by the at least one processor individually or collectively, cause the electronic device to: based at least in part on the first user input, display, via the display and concurrently with the first image, the first UI object, and the third UI object, a fourth UI object including a thumbnail image indicative of the first image as no filter is applied to the first image.
6. The electronic device of claim 1, wherein the instructions, when executed by the at least one processor individually or collectively, cause the electronic device to: after the first filter is generated, display, via the display and concurrently with the third image and the second UI object, a third UI object corresponding to a second filter generated previously before the first filter.
7. The electronic device of claim 6, wherein the third UI object includes an image and a text, and wherein each of the image and the text is indicative of the second filter.
8. The electronic device of claim 6, wherein the instructions, when executed by the at least one processor individually or collectively, cause the electronic device to: based at least in part on the first user input, display, via the display and concurrently with the first image, the first UI object, and the third UI object, a fourth UI object including a thumbnail image indicative of the first image as no filter is applied to the first image.
9. The electronic device of claim 1, wherein the instructions, when executed by the at least one processor individually or collectively, cause the electronic device to: as part of the generating of the first filter, store data corresponding to the first filter independently of the second image.
10. The electronic device of claim 1, wherein the instructions, when executed by the at least one processor individually or collectively, cause the electronic device to: as part of the displaying of the third image, replace a first property of at least a portion of the first image with a second property

based at least in part on a third property of at least a portion of the second image.

11. The electronic device of claim 10, wherein the first property includes a first color and the second property includes a second color different from the first color.

12. The electronic device of claim 1, further comprising: a camera, wherein the instructions, when executed by the at least one processor individually or collectively, further cause the electronic device to: identify, as the first image, an image captured via the camera.

13. The electronic device of claim 1, further comprising: a camera, wherein instructions, when executed by the at least one processor individually or collectively, further cause the electronic device to: identify, as the second image, an image captured via the camera.

14. The electronic device of claim 1, wherein the instructions, when executed by the at least one processor individually or collectively, further cause the electronic device to: after the first filter is generated, display, via the display, a fourth image, based at least in part on a fourth user input for editing the fourth image while the fourth image is displayed, display, via the display and concurrently with the fourth image, the first UI object, the second UI object, and a third UI object corresponding to a second filter generated previously before the first filter, based at least in part on a fifth user input for the second UI object, display, via the display, a fifth image generated by applying the first filter to the fourth image, and based at least in part on a sixth user input for the third UI object, display, via the display, a sixth image generated by applying the second filter to the fourth image.

15. The electronic device of claim 1, wherein the instructions, when executed by the at least one processor individually or collectively, further cause the electronic device to: store the third image in the memory.

16. The electronic device of claim 1, wherein the instructions, when executed by the at least one processor individually or collectively, further cause the electronic device to: display, via the display, an interface for adjusting a strength of the first filter to be applied to the first image.

17. The electronic device of claim 1, wherein the each of the first image and the third image includes a still image or a moving image.

18. The electronic device of claim 1, wherein the instructions, when executed by the at least one processor individually or collectively, cause the electronic device to: based on a gallery application being executed, display, via the display, the first image.

19. A method performed by an electronic device, the method comprising: displaying, via a display of the electronic device, a first image; receiving a first user input for editing the first image while the first image is displayed via the display; based at least in part on the first user input, displaying, via the display and concurrently with the first image, a first user interface (UI) object for generating a new filter; based at least in part on a second user input with respect to the first UI object, displaying, via the display, a plurality of images previously stored in memory of the electronic device; based at least in part on a third user input with respect to a second image among the plurality of images, generating, as the new filter, a filter corresponding to the second image; based on the generating of the filter, displaying, via the display, a third image and a second UI object, the third image corresponding to the first image, the second UI object corresponding to the filter, the third image being generated by applying the filter to the first image, the second UI object including a thumbnail image indicative of the second image; and storing the filter in the memory.

20. A non-transitory computer-readable medium having recorded thereon computer executable instructions, the computer executable instructions that, when executed by at least one processor of an electronic device individually or collectively, cause the electronic device to: display, via a display of the electronic device, a first image; receive a first user input for editing the first image while the first image is displayed via the display; based at least in part on the first user input, display, via the display and concurrently with the first image, a first user interface (UI) object for generating a new filter; based at least in part on a second user input with respect to the first UI object, display, via the display, a plurality of images previously stored in memory of the electronic

device; based at least in part on a third user input with respect to a second image among the plurality of images, generate, as the new filter, a filter corresponding to the second image; based on the generating of the filter, display, via the display, a third image and a second UI object, the third image corresponding to the first image, the second UI object corresponding to the filter, the third image being generated by applying the filter to the first image, the second UI object including a thumbnail image indicative of the second image; and store the filter in memory.
